Information Theory Mn908 and MICAS

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Exercise 3 of 7 October 2026

(solutions are due on 14 October 2026, 8h30)

Problem 1 ***On Fano's Inequality***

Let X take value in the set $\mathcal{X} = \{1, \ldots, m\}$, and let $p_i = \Pr[X = i]$ for $i = 1, \ldots, m$. Assume $p_1 \geq p_2 \geq \cdots \geq p_m$.

1. Suppose that you must guess X. Which guess would have the smallest probability of error? What is the probability of error P_e^* associated with this guess?
2. Maximize $H(X)$ over all choices of $p_1, p_2, \ldots, p_m$ for which the best guess has a probability of error P_e^*.

Problem 2 ***Markov Process and Entropy Rate***

Let $\{X_k\}_{k=1}^{\infty}$ be the following homogeneous (time-invariant) Markov process. For each positive integer $k > 1$, if $X_{k-1} = 0$, then $X_k = 0$ with probability $1/3$ and $X_k = 1$ with probability $2/3$, and if $X_{k-1} = 1$, then $X_k = 0$ with probability $1/2$ and $X_k = 1$ also with probability $1/2$.

a) Find the entropy rate $H(\{X_k\})$.

b) For each positive integer k, define $V_k = X_{2k}$. Is the process $\{V_k\}$ Markov? Be sure to justify your answer!

c) Which of the following two statements is true: $H(\{V_k\}) \geq H(\{X_k\})$ or $H(\{V_k\}) \leq H(\{X_k\})$?

d) For each positive integer k, define $Y_k = X_{k+1}$ if k is odd and $Y_k = X_{k-1}$ if k is even. Is the process $\{Y_k\}$ a Markov process? Be sure to justify your answer!

e) Find the entropy rate $H(\{Y_k\})$.

Problem 3 ***Window-Based Lempel-Ziv***

1. Encode the following sequences with the window-based Lempel-Ziv algorithm seen in class with window size 4 and using the simpler version of the unbounded integer encoding to encode the parameter L:
   - `1100000001111100111111110000`
   - `00000111011100111`

   What is your compression rate?
2. Decode the following sequence that has been encoded with the window-based Lempel-Ziv algorithm with window-size 8:
   - `0110001111101000000110011111010000`

Problem 4 ***Shannon–Fano Coding***

A discrete source X takes values in the alphabet $\mathcal{X} = \{a, b, c, d, e, f\}$ with probabilities

$$P_X = (0.30,\ 0.25,\ 0.20,\ 0.10,\ 0.10,\ 0.05).$$

a) Construct a binary Shannon–Fano code for this source by ordering the symbols according to decreasing probability and recursively splitting each group into two subgroups whose total probabilities are as close as possible. At each split, assign bit 0 to the first subgroup and bit 1 to the second.

b) Give the resulting codeword lengths and compute the average codeword length

$$\bar{L} = \sum_{x \in \mathcal{X}} P_X(x)\, \ell(x).$$

c) Compute the entropy $H(X)$ and compare it with $\bar{L}$. What is the coding efficiency $H(X)/\bar{L}$?

d) How many bits per source symbol would a fixed-length binary code require for the same alphabet? Compare this with the Shannon–Fano code.

e) Use your Shannon–Fano code to encode the source sequence `abacafed`. How many bits does the encoded sequence contain?